

Mithin Sagar S

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EDUCATION

Vellore Institute of Technology, Chennai

B.Tech, Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Aug 2023 – May 2027

Chennai, India

EXPERIENCE

Machine Learning Intern — Deep Learning & Computer Vision

Indira Gandhi Centre for Atomic Research (IGCAR)

May 2025 – Jun 2025

Kalpakkam, India

- Fine-tuned a **YOLOv8l** detector on high-resolution weld imagery for automated defect localization, curating the dataset and training 100 epochs on GPU.
- Evaluated with precision, recall, F1, confusion matrices and mAP — **99.1% mAP on training vs. 74.3% on held-out test** and analyzed the generalization gap to diagnose overfitting and drive dataset and split decisions.
- Automated the workflow end-to-end (preprocessing → inference → reporting), replacing a manual process and cutting memory footprint for stable runs on constrained hardware.

PROJECTS

Adversarial Robustness of Explainable AI — Attack & Defense Framework | *PyTorch, SHAP, LIME, Captum,*

XGBoost

2026

- Showed post-hoc explanations can be adversarially manipulated while predictions stay fixed, implementing **four attacks** (top-feature, prediction-stable, FGSM gradient-guided, targeted Integrated-Gradients) over three cybersecurity datasets totaling **246K+ samples**.
- Built PyTorch MLPs (BatchNorm/Dropout), XGBoost and logistic-regression baselines explained via SHAP, LIME, Integrated Gradients and permutation importance; an episodic Prototypical Network showed few-shot models drift **40–100% more**.
- Designed **four defenses** with custom composite losses (noise-stability, adversarial, explanation-regularization, hybrid); the hybrid cut mean explanation drift **91.1%** (0.0388 → 0.0035) with no classification loss.

EXAI-ResumeIntel — Explainable Resume-to-Role Matching at Scale | *SBERT, FastAPI, Docker, NumPy*

2026

- Engineered semantic matching over **1M+ job postings** by streaming a **4.96 GB memory-mapped float32 array** of 768-dim SBERT embeddings in chunks, keeping similarity search memory-bounded instead of loading the corpus into RAM.
- Implemented explainability from first principles: exact Shapley values for ≤ 12 features with 512-sample Kernel SHAP fallback, LIME over 300 perturbations, ranked counterfactuals, and a 22-skill/346-alien ontology using n-gram matching with 0.8-decay parent propagation.
- Refactored a monolithic explainer into six modules behind a coordinator, rebuilt the API on typed **FastAPI/Pydantic**, and deployed via Docker, Kubernetes and Nginx with 6 GB of artifacts auto-fetched at runtime.

MediXplain — Calibrated Disease Prediction with RAG Evidence Retrieval | *scikit-learn, FAISS, Flask, CI/CD*

2026

- Trained and cross-validated **seven** scikit-learn classifiers on a 4,920-record, 133-feature, 41-class dataset, wrapping estimators in CalibratedClassifierCV for calibrated confidence; best model reached **83.09%** 5-fold CV accuracy.
- Built RAG evidence retrieval over a **FAISS** inner-product index of L2-normalized 384-dim MiniLM embeddings for sub-millisecond top-K lookup with TF-IDF cosine fallback; shipped on Flask with GitHub Actions running flake8, black and 12 pytest tests on Python 3.10–3.12.

PUBLICATIONS

Automated AWS Resource Cleanup for Optimization of Cost and Security — *Proc. 2nd Intl. Conf. on Advanced*

Nexus of Data and Information Technology (ICANDIT 2026), INTI International University, Malaysia. Event-driven

Boto3/Lambda/EventBridge system inventorying EC2, EBS, S3, IAM and RDS across eight paginated service managers to detect and safely reclaim idle resources.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL

AI/ML: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, YOLOv8, OpenCV, SHAP, LIME, Captum, sentence-transformers (SBERT), FAISS, NumPy, pandas

Cloud & Infrastructure: AWS (Lambda, EventBridge, EC2, S3, EBS, IAM, CloudWatch, SNS), Boto3, Docker, Linux

Engineering: REST APIs, FastAPI, Flask, Git/GitHub, GitHub Actions (CI/CD), pytest, unittest, flake8, black

ACHIEVEMENTS & CERTIFICATIONS

Outreach Head, TechnoVIT & Vibrance, VIT Chennai (2024–2025) — led a 25+ member team across sponsorship, publicity and media for two flagship university festivals. **First Runner-Up**, Figma × Apple Vision Pro Design Challenge, GDSC VIT Chennai (2023).

Certifications: Introduction to Generative AI (Google Cloud); Python Data Structures and Programming for Everybody (Univ. of Michigan, Coursera); Google UX Design Professional Certificate.